Optimal reconstruction for an analytically solvable Euler family

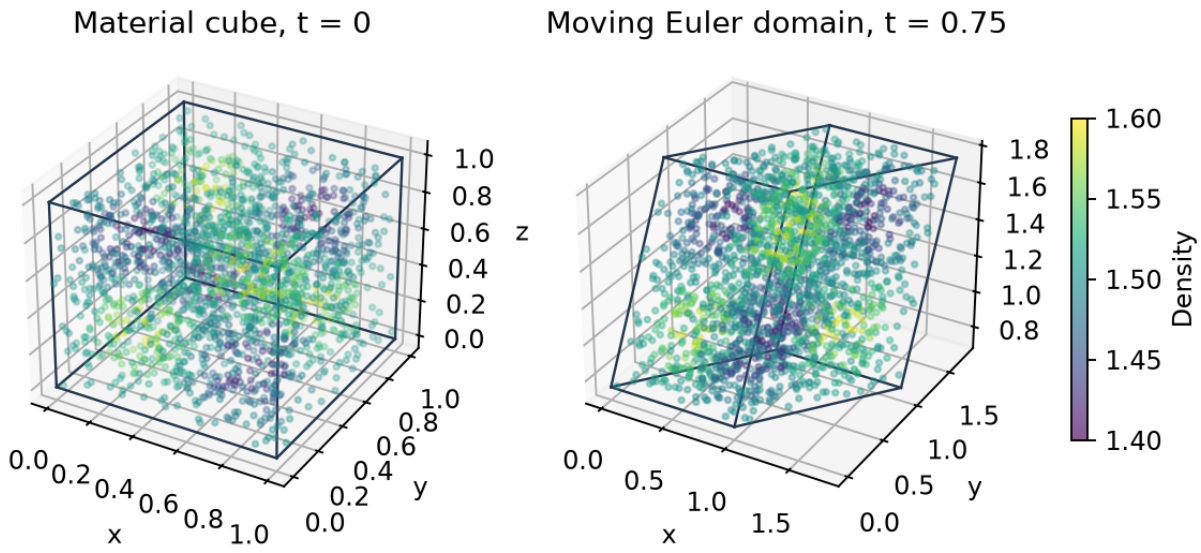
Simjecture research experiment • scope corrected 2 November 2026 • PR #90

Research status correction. This is a proof-workflow and reconstruction exercise. Its motion is already analytic, so it does not meet the requested goal of finding an optimal solver for unknown nonlinear Euler dynamics. The replacement 3D compressible Taylor–Green family is documented in demos/lean_euler/nonlinear; its accuracy and optimality remain unproved.

Result. A finite characteristic reconstruction attains a prescribed uniform physical-field error with work proportional to $(L/\varepsilon)^3$. Every deterministic adaptive point-query algorithm solving the same family has a matching worst-case lower bound. Lean checks the actual Euler equations, the approximation guarantee, and the work argument.

$$W^*(L, \varepsilon) = \Theta((L/\varepsilon)^3)$$

Scope. Full compressible Euler in three dimensions, $\gamma = 2$, with arbitrary C1 Lipschitz initial density in [1,2], a fixed affine shear velocity, constant pressure, and a prescribed moving material wall. The output error covers all five conserved variables throughout the moving cube.



The density shown is 1.5 + 0.1 sin(2πX) sin(2πY) sin(2πZ). The theorem covers the entire stated density family, not only this demonstration.

Meaning of optimal. The exponent is optimal up to fixed multiplicative constants in an exact-real RAM/oracle model. This is not a proof of the fastest wall time, the best constant, or optimality for pressure-driven or shock-forming Euler flows.

The problem that was frozen before proof construction

Material position $a = (X,Y,Z)$ lies in $Q = [0,1]^3$. The Eulerian domain is $F_t(Q)$, and the wall moves with F_t. The original contract uses $0 \le t \le 1$. The formulas and checked estimates also hold at any real t.

$$F_t(X,Y,Z) = (X + tY,\ Y + tZ,\ Z + t)$$
$$F_t^{-1}(x,y,z) = (x - ty + t^2z - t^3,\ y - tz + t^2,\ z - t)$$

$$\rho(t,x) = \rho_0(F_t^{-1}(x))$$
$$v(t,x) = (y - tz + t^2,\ z - t,\ 1), \quad p = 1$$
$$U = (\rho,\ \rho v_x,\ \rho v_y,\ \rho v_z,\ 1 + \rho|v|^2/2)$$

All variables are nondimensional. The input function $\rho_0: R^3 \to R$ is globally C1, takes values in [1,2], and has Lipschitz constant at most $L \ge 1$ in the coordinate maximum norm. It may vary independently in all three spatial coordinates. A competitor can learn it only by point evaluations.

Checked physical property	What it establishes
C1 regularity and canonical Euler residual	Actual time and spatial differentiation of all five conservation equations; ideal-gas pressure is recovered from the conserved state.
Initial condition	The field at $t = 0$ has density ρ_0, velocity $(Y,Z,1)$, and pressure 1.
Boundary motion	The material trajectory derivative equals the fluid velocity, including on every wall face.
Invertible trajectory map	Mapping an Eulerian evaluation point back to the material cube is exact in the real-arithmetic theorem.

This velocity has nonzero vorticity and nonzero convective terms, but its motion is known analytically and does not depend on ρ_0. The numerical problem is therefore recovery and transport of density within an exactly reducible Euler family. No general pressure dynamics are solved by this construction.

Accuracy is relative to the explicitly constructed classical solution. Uniqueness among every possible Euler solution is not a formalized claim.

Constructive accuracy and complete modeled work

1. Set n = ceil(3L/(4ε)).
2. Enumerate all n^3 material cells and query ρ_0 at their centers.
3. Retain the density table and the fixed reconstruction rule.
4. At an output point, use F_t^{-1}, its material cell, and the exact velocity to reconstruct U.

Every material point is within 1/(2n) of its cell center. Consequently:
density error ≤ L/(2n), and **full-state error ≤ 3L/(4n) ≤ ε**. The factor 3/2 comes from the energy component; the material velocity is (Y,Z,1). This is a uniform bound over the continuous domain.

Counted part	RAM work
Enumerating triples in three nested loops	$5n^3 + 4n^2 + 4n + 2$
Center arithmetic, oracle calls, table writes and fill-loop control	$13n^3 + 2$
Table initialization, resolution, casts and output descriptor	$n^3 + 8$
Total	$19n^3 + 4n^2 + 4n + 12 \le 39n^3$

For the stated small-tolerance regime, the checked upper bound is **$W \le 312(L/\varepsilon)^3$**. The RAM model charges scalar casts, additions, divisions, ceiling, indexing/storage, and oracle calls at unit cost. The table has n^3 values; index enumeration and storage are included.

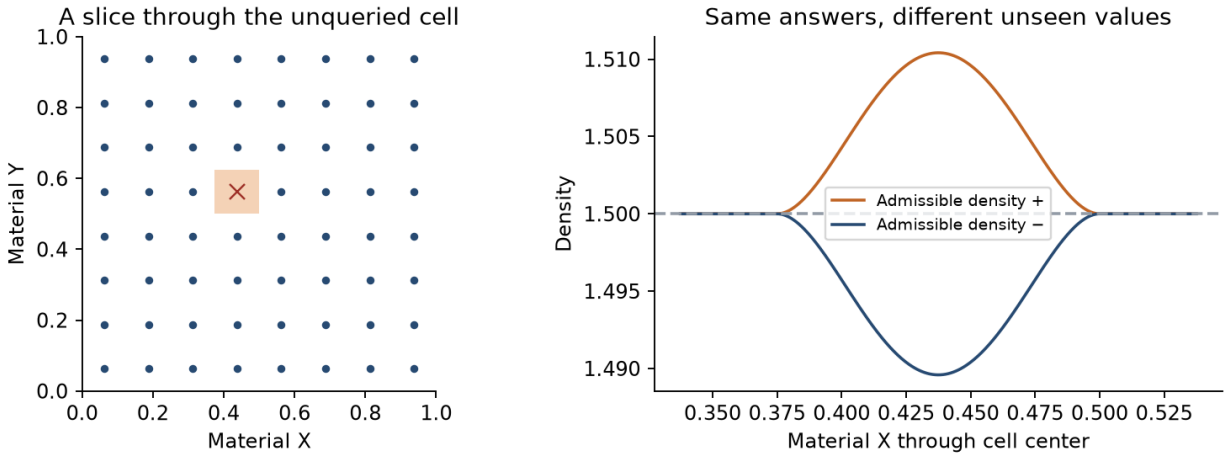
The output is a finite representation of the full field. Evaluating additional requested points has additional cost. Finite functions represent constant-time RAM arrays in the formal program; the result is not a cost bound for persistent function closures. Bit complexity, floating-point rounding, compiler behavior and hardware scheduling are outside this model.

The NumPy demonstration implements the same mathematical reconstruction with vectorized arrays. Lean has not verified that Python implementation. Its observed elapsed time is reported separately from the formal RAM count.

A lower bound against adaptive competitors

The machine model permits arbitrary internal state, adaptive queries anywhere in R^3, and arbitrary final output representations. Only admissible inputs must terminate. A point query costs one unit; any additional scalar or memory work is nonnegative.

Follow the transcript produced by the constant input $\rho_0 = 3/2$. If fewer than n^3 queries are made, one of n^3 disjoint small cells receives no informative query. Lean constructs two globally smooth densities, $3/2 \pm b$, supported around that untouched cell. Their bounds and Lipschitz constants are checked. Both give the same answers throughout the transcript but differ by 4ε at the cell center. Their common output cannot be within ε of both.



The figure uses a concrete quartic bump for numerical illustration. The proof constructs a tensor product of Mathlib smooth one-dimensional bumps and proves a fixed Lipschitz bound exists.

For one fixed $K \geq 1$, independent of L and ε, every correct competitor satisfies

$$(L/(4K\varepsilon))^3 \leq \text{number of queries} \leq \text{total charged work},$$

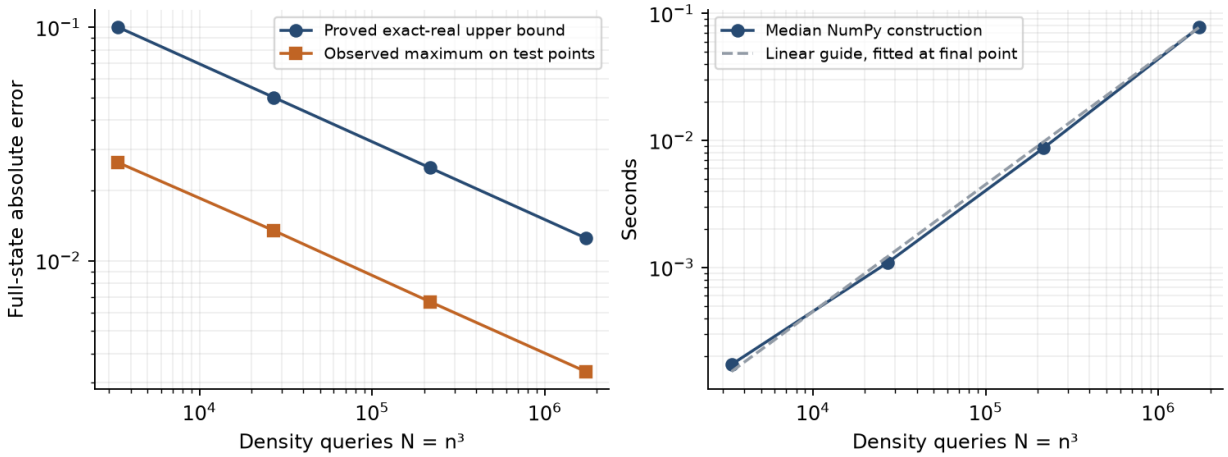
provided $L \geq 1$ and $0 < \varepsilon \leq 1/(4K)$.

K is not numerically evaluated. The theorem establishes the matching asymptotic exponent. It does not supply a sharp finite-tolerance lower-bound constant or a measured speedup. The lower bound concerns the same full-state accuracy target because density is its first component.

Local numerical demonstration and negative controls

The example uses the three-coordinate sinusoidal density on page 1, with L = 2. Each grid was constructed three times. Evaluation used 100,008 material points at t = 0, 0.25, 0.75 and 1, including cube corners. Construction timings include sample coordinates, density calls, table allocation and descriptor creation; reference/error analysis is separate.

ε	n	Queries	Proved bound	Observed max	Build, ms
0.1	15	3,375	0.1000	0.02633	0.173
0.05	30	27,000	0.0500	0.01345	1.090
0.025	60	216,000	0.0250	0.00668	8.700
0.0125	120	1,728,000	0.0125	0.00335	77.850



Accuracy negative control. Replacing the required n = 30 by n = 10 fails on a smooth tanh density with global Lipschitz constant 2. At the recorded point, full-state error rises from 0.04856 to 0.14795, above ε = 0.05. The corresponding modified Lean source also fails to check.

The independent central-difference Euler residual check gave a maximum of 1.55e-09 with step 10^{-5}. This is a numerical consistency check; the zero residual is proved separately in Lean.

These timings are local observations for a simple density oracle. No comparison is made to Rusanov/MUSCL runs with different periodic boundaries, and no general-purpose solver speedup is inferred.

Audit, reproduction, and what remains outside the result

20 named statements passed compilation and a separate transitive axiom audit. Allowed dependencies are propext, Classical.choice and Quot.sound. The checker retains the exact source, commands, runtime identity, diagnostics and hashes. The changed-resolution source is retained as an unsuccessful negative control.

Central statement: **Machines.certified_general_optimality**.
Physical solution: **Certificate.admissible_is_classical_euler**.
Algorithm/value connection: **Work.solver_program_matches_query_algorithm**.

From the Simjecture repository root:

```
.venv/bin/simjecture install lean
.venv/bin/python demos/lean_euler/optimality/run_local.py \
  --output artifacts/euler-optimality-replay
```

Use a new output directory. The replay creates an isolated local worker, runs the numerical experiment and bounded Lean capability, and retains lab receipts. The proof worker reserves 8 GB of address space; Lean itself uses one thread and a 2 GB managed-memory setting. A 4 GB address-space replay failed while loading the analysis libraries, before proof diagnostics. No global memory limit was changed.

Source SHA-256: 671378957588a6d496cd9fb8aae3df22415700c8a65aa98a83bd8f59f56736b9

Established	Outside this proof
Classical Euler equations for the frozen shear family	General pressure-driven flow, shocks, turbulence, other boundary/initial classes
Uniform full-state accuracy for the returned representation	Certified Python/IEEE error, input-class verification from samples
Matching worst-case work scaling against adaptive point-oracle machines	Best constants, randomized algorithms, derivative/integral/formula oracles, fastest hardware runtime

Function recovery lower bounds have a substantial prior literature: C. A. Micchelli, T. J. Rivlin and S. Winograd, The optimal recovery of smooth functions (1976). Affine Euler motions also predate this experiment; see T. C. Sideris, Global existence and asymptotic behavior of affine motion of 3D ideal fluids surrounded by vacuum, whose boundary/data family differs from this one.

This PR makes a scoped proof workflow reproducible. Mathematical novelty and priority are not established. A next research step would require a new frozen contract with unknown pressure dynamics, a proved numerical accuracy bound, and a lower bound for that enlarged class.